

Computer Programming 143 – Lecture 25

File Processing II

Faculty of Engineering



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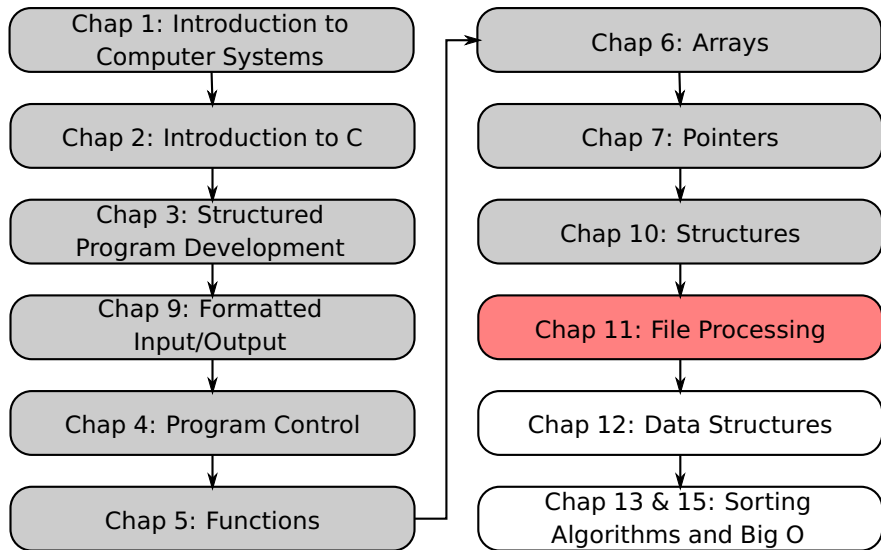
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Module Overview



Lecture Overview

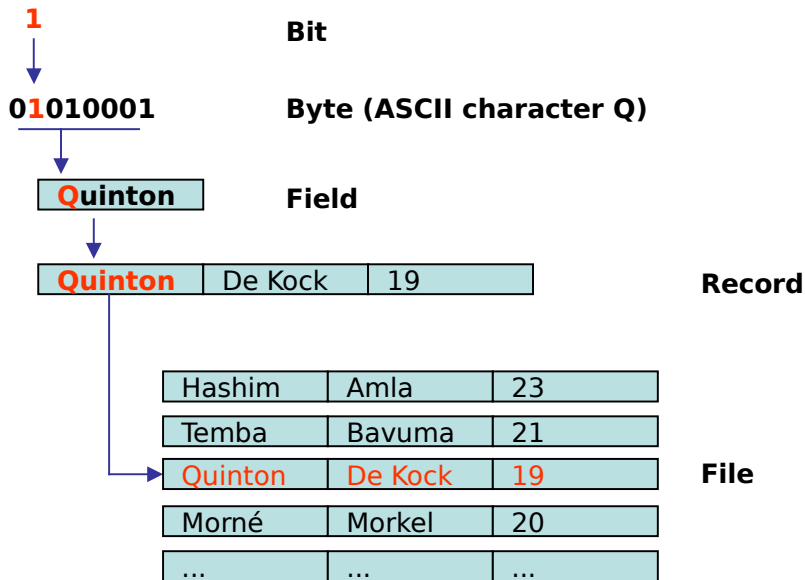
- 1 The Data Hierarchy (1.3)
- 2 Write to/Read from Binary File (11.6-11.8)

1.3 The Data Hierarchy I

Data Hierarchy

- **Bit** – smallest data item
 - Value of 0 or 1
- **Byte** – 8 bits
 - Used to store a character
 - Decimal digits, letters, and special symbols
- **Field** – group of characters conveying meaning
 - Example: your name
- **Record** – group of related fields
 - Represented by a struct or a **class**
 - Example: In a payroll system, a record for a particular employee that contained his/her identification number, name, address, etc.
- **File** – group of related records
 - Example: payroll file
- **Database** – group of related files

1.3 The Data Hierarchy II



1.3 The Data Hierarchy III

Data files

- Record key
 - Identifies a record to facilitate the retrieval of specific records from a file (eg. ID)

ID	NAME	SURNAME
253	Hashim	Amla
254	Temba	Bavuma
255	Quinton	De Kock
256	Morné	Morkel
257	...	...

- Sequential file
 - Records typically sorted by key

11.6 Write to/Read from Binary File I

Unformatted I/O functions

- `fwrite`
 - Transfer bytes from a location in memory to a file
- `fread`
 - Transfer bytes from a file to a location in memory

- Example:

```
fwrite( &number, sizeof( int ), 1, fPtr );
```

- `&number` – Location to transfer bytes from
- `sizeof( int )` – Number of bytes to transfer
- `1` – For arrays, number of elements to transfer. In this case, “one element” of an array is being transferred
- `fPtr` – File to transfer to or from

11.6 Write to/Read from Binary File II

Writing structures

```
fwrite( &myObj, sizeof( struct myStrct ), 1, fPtr );
```

- `sizeof` – returns size in bytes of object in parentheses

Reading structures

```
fread( &myObj, sizeof( struct myStrct ), 1, fPtr );
```

- `sizeof` – returns size in bytes of object in parentheses

Writing and reading array of structures

```
fwrite( myArr, sizeof( struct myStrct ), numElements, fPtr );  
fread( myArr, sizeof( struct myStrct ), numElements, fPtr );
```

- `numElements` – is the number of array elements

11.7 Sequential Write to Binary File I

```
/* Example of writing sequentially to a binary file */
#include <stdio.h>
#include <stdlib.h>

struct coord { // structure to store coordinates
    double x;
    double y;
}; // end structure coord
typedef struct coord Coord;

int main( void )
{
    char filename[ 20 ]; // string to store filename
    FILE *cfPtr;          // file pointer
    int i;                 // counter
    Coord vertex;          // coordinates of a vertex of triangle

    printf( "Enter file name: " );
```

11.7 Sequential Write to Binary File II

```
scanf( "%19s", filename ); // reads filename from user
cfPtr = fopen( filename, "wb" ); // opens binary file for writing
if ( cfPtr == NULL ) { // exits if file cannot be opened
    printf( "ERROR: File could not be opened!\n" );
}
else {
    printf( "\nPlease specify the vertices of the triangle\n" );
    for ( i = 1; i <= 3; i++ ) {
        printf( "Enter the coordinates of vertex %d: ", i );
        // reads a vertex from user
        scanf( "%lf%lf", &vertex.x, &vertex.y );
        // writes vertex to file
        fwrite( &vertex, sizeof( Coord ), 1, cfPtr );
    }
    fclose( cfPtr ); // closes file
}
return 0; // indicates successful termination
} // end main
```

11.8 Sequential Read from Binary File I

```
/* Example of reading sequentially from a binary file */
#include <stdio.h>
#include <math.h>
#include <stdlib.h>

struct coord { // structure to store coordinates
    double x;
    double y;
}; // end structure coord
typedef struct coord Coord;

int main( void ) {
    char filename[20]; // string to store filename
    FILE *cfPtr;       // file pointer
    int i;             // counter
    Coord vertices[3]; // array for triangle coordinates
```

11.8 Sequential Read from Binary File II

```
printf( "Please enter filename: " );
scanf( "%s", filename ); // reads filename from user
cfPtr = fopen( filename, "rb" ); // opens file for reading
if ( cfPtr == NULL ) { // exits if file cannot be opened
    printf( "ERROR - file could not be opened!\n" );
}
else {
    printf( "\n" );
    for ( i = 0; i < 3; i++ ) {
        // reads one set of coordinates
        fread( &vertices[i], sizeof( Coord ), 1, cfPtr );
        printf( "Vertex %d: ( %.11f, %.11f )\n", i + 1,
            vertices[i].x, vertices[i].y );
    }
    fclose( cfPtr ); // closes file
}
return 0; // indicates successful termination
} // end main
```

Today

File Processing II

- Data hierarchy
- Binary files

Next lecture

File Processing III

- Random-access files

Homework

- 1 Study Sections 1.3, 11.5-11.8 in Deitel & Deitel
- 2 Do Self Review Exercises 11.4 in Deitel & Deitel
- 3 Do Exercises 11.5, 11.10 in Deitel & Deitel